

inclusion-minimal: A set that satisfies a property, but no proper subset satisfies that property

Recall Inductively Defined Structures

1. The set $\mathbb{N}$ of natural numbers: the **inclusion-minimal** set S such that

1. $0 \in S$
2. If $x \in S$, $x+1 \in S$

2. The set Σ^* of all strings over Σ : the **inclusion-minimal** set S such that

1. $\epsilon \in S$
2. If $u \in S$ and $a \in \Sigma$, then $au \in S$

3. The set $\mathcal{B}[A]$ of all binary trees over A : the **inclusion-minimal** set S such that

1. $\emptyset \in S$
2. If $l, r \in S$ and $a \in A$, then $\text{tree}(a; l, r) \in S$

4. The set $\text{List}[A]$ of all lists over A : the **inclusion-minimal** set S such that

1. $[] \in S$
2. If $a \in A$, and $l \in \text{List}[A]$, then $[a] :: l \in S$ → prepend operation

5. The set $\mathcal{PL}$ of all well-formed formulas in propositional logic: the **inclusion-minimal** set S such that

1. Statement letters $P, Q, \dots \in S$
2. If $P, Q \in S$, then $P \vee Q \in S$
3. If $P \in S$, then $\neg P \in S$

6. The set $\mathcal{T}[A]$ of all trees over A : the **inclusion-minimal** set S such that

1. $\emptyset \in S$
2. If $T \in \text{List}[S]$, and $a \in A$, then $\text{Tree}(a; T) \in S$

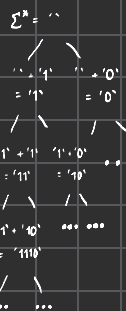
Formal Languages

The set Σ^* of all strings over Σ

"Consider a finite set Σ called an alphabet (set), the set Σ^* of all strings over Σ , are inductively defined as defined as the inclusion-minimal set such that

1. The empty string $\epsilon \in \Sigma^*$ (base case)
2. If $u \in \Sigma^*$ and $a \in \Sigma$, then $au \in \Sigma^*$

Ex. let $\Sigma = \{0, 1\}$



Ex. let $\Sigma = \{0\}$

$$\Sigma^* = \{\epsilon, 0, 00, 000, \dots\}$$

Ex. let $\Sigma = \emptyset$

$$\Sigma^* = \{\epsilon\}$$

Principle of Induction for Σ^*

Let $P: \Sigma^* \rightarrow \mathbb{B} = \{\text{True}, \text{False}\}$ be a predicate. The predicate P holds over Σ^* iff the following are true:

1. The base case $P(\epsilon)$ is true
2. The inductive step: If $P(u)$ is true and $a \in \Sigma$, then $P(au)$ is true, is true

Formal Languages

The length $|w|$ of a string w is defined recursively as

$$|w| \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } w = \epsilon, \\ 1 + |u| & \text{if } w = au \text{ (otherwise)} \end{cases}$$

ex. 1001101

$$\begin{aligned} &= 1 + |011101| \\ &= 1 + 1 + |11101| \\ &= 1 + 1 + 1 + |1101| \\ &= 1 + 1 + 1 + 1 + |01| \\ &= 1 + 1 + 1 + 1 + 1 + |1| \\ &= 1 + 1 + 1 + 1 + 1 + 0 = 5 \end{aligned}$$

ex. let $\Sigma = \{a, b, c\}$

$$\begin{aligned} \text{cab} \cdot \text{bage} &= c(ab \cdot \text{bage}) \\ &= c(a(b \cdot \text{bage})) \\ &= c(a(b(c \cdot \text{bage}))) \\ &= c(a(b(\text{bage}))) \\ &= c(a(\text{bbage})) = c(\text{abbbage}) \\ &= \text{cabbage} \end{aligned}$$

The concatenation $w \cdot v$ of $w, v \in \Sigma^*$ is defined as

$$w \cdot v \stackrel{\text{def}}{=} \begin{cases} v & \text{if } w = \epsilon, \\ a(u \cdot v) & \text{if } w = au \text{ (otherwise)} \end{cases}$$

Example

For all $u, v \in \Sigma^*$, $|w \cdot v| = |w| + |v|$

Proof:

Fix an arbitrary $v \in \Sigma^*$ and let $P(u): |u \cdot v| = |u| + |v|$

Base Case: when $w = \epsilon$

$$|w \cdot v| = |\epsilon \cdot v| = |v| = |w| + 0 = |v| + |\epsilon| = |v| + |w|$$

Inductive Step: when $w = au \Rightarrow |w| = 1 + |u|$

$$|w \cdot v| = |au \cdot v| = |a(u \cdot v)| = 1 + |u \cdot v| \stackrel{\text{IH}}{=} 1 + |u| + |v| = |w| + |v|$$

Conclusion

By the principle of mathematical induction, the statement is true

The reversal $\text{rev}(w)$ of $w \in \Sigma^*$ is defined as

$$\text{rev}(w) \stackrel{\text{def}}{=} \begin{cases} \epsilon & \text{if } w = \epsilon \\ \text{rev}(wa) & \text{if (otherwise) } w = wa \end{cases}$$

$$\text{ex. } \text{rev}(xxzy) = \text{rev}(xzy)x = \text{rev}(zy)xx = \text{rev}(y)zxx = \text{rev}(\epsilon)yzxx = \epsilon yzxx = yzxx$$

Formal Languages

Language: A (formal) language L over Σ is any subset of Σ^*

$$\text{ex. if } \Sigma = \{0, 1\}, \text{ then } L_1 = \{0, 1\} \subseteq \Sigma^*, L_2 = \{00, 11\} \subseteq \Sigma^*$$

Given languages $K, L \subseteq \Sigma^*$, their union $K \cup L$ is the set of all strings from K

together with all the strings from L

$$K \cup L \stackrel{\text{def}}{=} \{w \in \Sigma^* \mid w \in K \text{ or } w \in L\}$$

Given languages $K, L \subseteq \Sigma^*$, the concatenation KL is the set of all concatenates of each string in

K with each string in L

$$KL \stackrel{\text{def}}{=} \{uv \in \Sigma^* \mid u \in K \text{ and } v \in L\}$$

Given a language $L \subseteq \Sigma^*$, and a natural number $k \in \mathbb{N}$, the k^{th} power L^k of the language L

is the set of all concatenates of length k , formed from words in L

$$L^k \stackrel{\text{def}}{=} \begin{cases} \{\epsilon\} & \text{if } k=0, \\ L^{k-1} & \text{if (otherwise) } k > 0 \end{cases}$$

Example

Given an alphabet Σ , and $k \in \mathbb{N}$, we have $\Sigma^k = \{w \in \Sigma^* \mid |w| = k\}$

Proof

$$\text{Let } P(k): \Sigma^k = \{w \in \Sigma^* \mid |w| = k\}$$

Base case ($k=0$)

$$\Sigma^0 = \{\epsilon\} \text{ by definition, as } \epsilon \text{ is (by definition) the only string of length 0}$$

$$\text{Therefore, } \Sigma^0 = \{w \in \Sigma^* \mid |w| = 0\} = \{\epsilon\}, \text{ so the base case holds}$$

Inductive Hypothesis

Assume that $\Sigma^{k-1} = \{w \in \Sigma^* \mid |w| = k-1\}$. We proceed via double subset inclusion

$$\subseteq \rightarrow \Sigma^k \subseteq \{w \in \Sigma^* \mid |w| = k\}$$

For any $w \in \Sigma^k$ we have $w = au$, where $a \in \Sigma^1$ and $u \in \Sigma^{k-1}$

Then we have

$$|w| = |au| = |a| + |u| = 1 + |u| = 1 + k - 1 = k$$

$$\text{Therefore, } w \in \{w \in \Sigma^* \mid |w| = k\}$$

Conclusion

Therefore, $P(0)$ is true and $P(k-1) \Rightarrow P(k)$, therefore by the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$

□